

Add Items in List

```
In [ ]: 1)append()  
        2)extend()  
        3)insert()
```

1.append()

```
In [ ]: It is used to add items at last index.  
        Syntax:-List.append(value)
```

```
In [1]: list1=[2,3,4,5,6]  
        list1.append(100)  
        list1
```

```
Out[1]: [2, 3, 4, 5, 6, 100]
```

```
In [2]: city_names=["Pune","Mumbai","Banglore","Delhi"]  
        city_names.append("Hydrabad")  
        city_names
```

```
Out[2]: ['Pune', 'Mumbai', 'Banglore', 'Delhi', 'Hydrabad']
```

```
In [3]: city_names=["Pune","Mumbai","Banglore","Delhi"]  
        city_names.append({'a':97,'z':85})  
        city_names
```

```
Out[3]: ['Pune', 'Mumbai', 'Banglore', 'Delhi', {'a': 97, 'z': 85}]
```

```
In [4]: list1=[2,3,4,5,6]  
        list1.append(200,300)  
        list1
```

TypeError

Traceback (most recent call last)

Cell In[4], line 2

```
1 list1=[2,3,4,5,6]  
----> 2 list1.append(200,300)  
      3 list1
```

TypeError: list.append() takes exactly one argument (2 given)

```
In [6]: list1=[2,3,4,5,6]  
        len(list1)
```

```
Out[6]: 5
```

2)extend()

```
In [ ]: List.extend(iterable)
        iterable>>List or tuple
```

```
In [7]: list1=[2,5,4,6,7]
        list2=[10,20,30,40]
        list1.append(list2)
        list1
```

```
Out[7]: [2, 5, 4, 6, 7, [10, 20, 30, 40]]
```

```
In [9]: list1=[2,5,4,6,7]
        list2=[10,20,30,40]
        list1.extend(list2)
        print(list1)
        print(list2)
```

```
[2, 5, 4, 6, 7, 10, 20, 30, 40]
[10, 20, 30, 40]
```

```
In [10]: list1=[2,5,4,6,7]
        list2=[10,20,30,40]
        list2.extend(list1)
        print(list1)
        print(list2)
```

```
[2, 5, 4, 6, 7]
[10, 20, 30, 40, 2, 5, 4, 6, 7]
```

```
In [11]: list1=[10,20,30,40]
        list1.extend([1,2,3,4])
        list1
```

```
Out[11]: [10, 20, 30, 40, 1, 2, 3, 4]
```

```
In [12]: list1=[1,2,3,4,5]
        list2=[10,20,30,40,50]
        list3=list1+list2
        print(list1)
        print(list2)
        print(list3)
```

```
[1, 2, 3, 4, 5]
[10, 20, 30, 40, 50]
[1, 2, 3, 4, 5, 10, 20, 30, 40, 50]
```

```
In [13]: x=[10,20,30,40,50]
        y=(8,6,4,2)
        x.extend(y)
        print(x)
        print(y)
```

```
[10, 20, 30, 40, 50, 8, 6, 4, 2]
(8, 6, 4, 2)
```

```
In [14]: x=[10,20,30,40,50]
y=(8,6,4,2)
y.extend(x)
print(x)
print(y)
```

AttributeError

Traceback (most recent call last)

Cell In[14], line 3

```
1 x=[10,20,30,40,50]
2 y=(8,6,4,2)
----> 3 y.extend(x)
4 print(x)
5 print(y)
```

AttributeError: 'tuple' object has no attribute 'extend'

```
In [16]: x=[10,20,30,40,50]
y=[8,7,6,5,4]
for i in y:
    x.append(i)
    print(x)
```

```
[10, 20, 30, 40, 50, 8]
[10, 20, 30, 40, 50, 8, 7]
[10, 20, 30, 40, 50, 8, 7, 6]
[10, 20, 30, 40, 50, 8, 7, 6, 5]
[10, 20, 30, 40, 50, 8, 7, 6, 5, 4]
```

3)insert()

Syntax: insert(index,value)

```
In [17]: x=[2,3,4,5,6]
x.insert(1,500)
print(x)
```

```
[2, 500, 3, 4, 5, 6]
```

```
In [18]: city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names.append("Hydrabad")
city_names
```

```
Out[18]: ['Pune', 'Banglore', 'Mumbai', 'Delhi', 'Hydrabad']
```

```
In [19]: city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names.insert(2,"Hydrabad")
city_names
```

```
Out[19]: ['Pune', 'Banglore', 'Hydrabad', 'Mumbai', 'Delhi']
```

```
In [26]: city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names.insert(2,"Hydrabad")
city_names
```

```
Out[26]: ['Pune', 'Banglore', 'Hydrabad', 'Mumbai', 'Delhi']
```

Delete Items From List

```
In [ ]: 1)remove()
2)pop()
3)clear()
```

1)remove()

```
In [ ]: Syntax:List.remove(Value)
```

```
In [27]: city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names.remove("Delhi")
city_names
```

```
Out[27]: ['Pune', 'Banglore', 'Mumbai']
```

```
In [28]: city_names=["Pune","Banglore","Mumbai","Delhi"]
city_names.remove("DELHI")
city_names
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[28], line 2
      1 city_names=["Pune","Banglore","Mumbai","Delhi"]
----> 2 city_names.remove("DELHI")
      3 city_names
```

```
ValueError: list.remove(x): x not in list
```

```
In [29]: city_names=["Pune","Banglore","Mumbai","Delhi"]
city="DELHI"
if city in city_names:
    city_names.remove(city)
    print("item deleted")
else:
    print("Item not found")
```

```
Item not found
```

2)pop():

```
In [ ]: List.pop(index)
```

```
In [31]: city_names=["Pune","Banglore","Mumbai","Delhi"]  
city_names.pop(2)  
city_names
```

```
Out[31]: ['Pune', 'Banglore', 'Delhi']
```

```
In [33]: list1=[10,20,30,40,50]  
list1.pop()  
list1
```

```
Out[33]: [10, 20, 30, 40]
```

```
In [34]: list1=[10,20,30,40,50]  
list1.pop(-1)  
list1
```

```
Out[34]: [10, 20, 30, 40]
```

```
In [35]: list1=[10,20,30,40,50]  
list1.pop(-2)  
list1
```

```
Out[35]: [10, 20, 30, 50]
```

3)clear()

```
In [ ]: List.clear()
```

```
In [36]: list1=[2,3,4,5]  
list1.clear()  
list1
```

```
Out[36]: []
```

del keyword

```
In [37]: city_names=["Pune","Banglore","Mumbai","Delhi"]  
city_names[0]
```

```
Out[37]: 'Pune'
```

```
In [38]: city_names=["Pune","Banglore","Mumbai","Delhi"]  
del city_names[1]  
city_names
```

```
Out[38]: ['Pune', 'Mumbai', 'Delhi']
```

7)index()

```
In [ ]: index=List.index(value)
```

```
In [39]: num=[20,30,40,50,60]
num.index(50)
```

Out[39]: 3

```
In [40]: num=[20,30,40,50,70,100]
n=40
if n in num:
    index=num.index(n)
    print(f"Index of {n} is {index}")
```

Index of 40 is 2

```
In [42]: num=[20,30,40,50,70,100]
n=440
if n in num:
    index=num.index(n)
    print(f"Index of {n} is {index}")
else:
    print(f"{n} is not present in list")
```

440 is not present in list

```
In [44]: list1=[2,3,'python',100,200,'machine','learning']
n=1000
if n in list1:
    index=list1.index(n)
    print(f"Index of {n} is {index}")
else:
    print(f"{n} is not present in list")
```

1000 is not present in list

```
In [45]: list1=[2,3,'python',100,200,'machine','learning']
n='python'
if n in list1:
    index=list1.index(n)
    print(f"Index of {n} is {index}")
else:
    print(f"{n} is not present in list")
```

Index of python is 2

8)count()

```
In [ ]: number_of_occurences=List.count(Value)
```

```
In [48]: list1=[2,3,'python',100,'Python',200,'machine','learning',20,'python','python']  
n='python'  
list1.count(n)
```

Out[48]: 3

```
In [49]: list1=[2,3,'python',100,'Python',200,'machine','learning',20,'python','python']  
n='Python'  
list1.count(n)
```

Out[49]: 1

9)copy()

```
In [50]: a=50  
b=a  
print(a)  
print(b)
```

50
50

```
In [51]: a=50  
b=a  
print(a)  
print(b)  
  
b=b+5  
print(a)  
print(b)
```

50
50
50
55

```
In [52]: x=[4,5,6,7,8]  
y=x  
  
print(x)  
print(y)
```

[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8]

```
In [53]: x=[4,5,6,7,8]
        y=x
```

```
print(x)
print(y)
```

```
x.append(100)
print(x)
print(y)
```

```
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8, 100]
[4, 5, 6, 7, 8, 100]
```

```
In [54]: x=[4,5,6,7,8]
        y=x #reference
```

```
print(x)
print(y)
```

```
y.append(200)
print(x)
print(y)
```

```
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8, 200]
[4, 5, 6, 7, 8, 200]
```

```
In [55]: x=[4,5,6,7,8]
        y=x.copy()
        print(x)
        print(y)
```

```
y.append(200)
print(x)
print(y)
```

```
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8]
[4, 5, 6, 7, 8, 200]
```

deep copy

```
In [1]: import copy
        x=[4,5,[6,7,8,10], 'python', 30,40]
        y=copy.deepcopy(x)
        x[2][0]=600
        print(x)
        print(y)
```

```
[4, 5, [600, 7, 8, 10], 'python', 30, 40]
[4, 5, [6, 7, 8, 10], 'python', 30, 40]
```


10)sort()

```
In [2]: list1=[4,2,6,7,8,1]
list1.sort()
list1
```

Out[2]: [1, 2, 4, 6, 7, 8]

```
In [3]: list1=[4,2,3,6,1,8]
list1.sort(reverse=False)
list1
```

Out[3]: [1, 2, 3, 4, 6, 8]

```
In [4]: list1=[4,2,3,6,1,8]
list1.sort(reverse=True)
list1
```

Out[4]: [8, 6, 4, 3, 2, 1]

```
In [6]: list2=['Hin','Bio','Eng','Math','Che']
list2.sort()
list2
```

Out[6]: ['Bio', 'Che', 'Eng', 'Hin', 'Math']

11)sorted

```
In [7]: x=[4,2,1,5,6,11,9]
y=sorted(x)
print(x)
print(y)
```

```
[4, 2, 1, 5, 6, 11, 9]
[1, 2, 4, 5, 6, 9, 11]
```

```
In [8]: x=[4,2,1,5,6,11,9]
y=sorted(x,reverse=True)
print(x)
print(y)
```

```
[4, 2, 1, 5, 6, 11, 9]
[11, 9, 6, 5, 4, 2, 1]
```

12)reverse()

```
In [9]: list1=[2,4,7,6,1,22]
list1[::-1]
```

Out[9]: [22, 1, 6, 7, 4, 2]

```
In [10]: list1=[3,5,62,1,6,4]
list1.reverse()
list1
```

```
Out[10]: [4, 6, 1, 62, 5, 3]
```

```
In [11]: list2=['Hin','Bio','Eng','Math','Che']
list2.reverse()
list2
```

```
Out[11]: ['Che', 'Math', 'Eng', 'Bio', 'Hin']
```

13)reversed()

```
In [13]: list2=['Hin','Bio','Eng','Math','Che']
new_list=list(reversed(list2))
print(new_list)
```

```
['Che', 'Math', 'Eng', 'Bio', 'Hin']
```

```
In [14]: list2=['Hin','Bio','Eng','Math','Che']
new_list=tuple(reversed(list2))
print(new_list)
```

```
('Che', 'Math', 'Eng', 'Bio', 'Hin')
```

```
In [15]: list2=['Hin','Bio','Eng','Math','Che']
new_list=set(reversed(list2))
print(new_list)
```

```
{'Hin', 'Che', 'Eng', 'Math', 'Bio'}
```

```
In [16]: list2=['Hin','Bio','Eng','Math','Che']
new_list=reversed(list2)
print(new_list)
```

```
<list_reverseiterator object at 0x0000022C20EE20E0>
```

Built In Function

```
In [ ]: 1)sorted()
2)reversed()
3)sum()
```

```
In [17]: list1=[10,20,30]
sum(list1)
```

```
Out[17]: 60
```

```
In [18]: list1=[10,20,30,'python']  
sum(list1)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[18], line 2  
      1 list1=[10,20,30,'python']  
----> 2 sum(list1)  
  
TypeError: unsupported operand type(s) for +: 'int' and 'str'
```

```
In [19]: '5'+'6'# We use + operator here for concatenation
```

```
Out[19]: '56'
```

```
In [20]: 5+6  #+ use for addition
```

```
Out[20]: 11
```

4)max()

```
In [21]: list1=[3,4,56,2,1]  
max(list1)
```

```
Out[21]: 56
```

5)min()

```
In [22]: list1=[3,4,56,2,1]  
min(list1)
```

```
Out[22]: 1
```

Nested List

```
In [23]: x=[[2,3,4],[5,6,7],[10,20,30],[100,300,500]]  
x
```

```
Out[23]: [[2, 3, 4], [5, 6, 7], [10, 20, 30], [100, 300, 500]]
```

```
In [24]: len(x)
```

```
Out[24]: 4
```

```
In [25]: x=[[2,3,4],[5,6,7],[10,20,30],[100,300,500]]  
x[0]
```

```
Out[25]: [2, 3, 4]
```

```
In [26]: x=[[2,3,4],[5,6,7],[10,20,30],[100,300,500]]  
x[-1]
```

Out[26]: [100, 300, 500]

```
In [27]: x=[[2,3,4],[5,6,7],[10,20,30],[100,300,500]]  
x[2][0]
```

Out[27]: 10

```
In [28]: x=[[2,3,4],[5,6,7],['python',20,30],[100,300,500]]  
x[2][0][2]
```

Out[28]: 't'

```
In [29]: x=[[2,3,4],[5,6,7],[[1,2,3],20,30],[100,300,500]]  
x[2][0][2]
```

Out[29]: 3

```
In [32]: x=[[2,3,4],[5,6,7],[[1,2,3],20,30],[100,300,500]]  
len(x)
```

Out[32]: 1

```
In [34]: x=[[2,3,4],[5,6,7],[[999,2,3],20,30],[100,300,500]]  
x[0][2][0][0]
```

Out[34]: 999

In []: